

## Supplementary Materials

# Differential coding of fruit, leaf and microbial odours in the brains of *Drosophila suzukii* and *Drosophila melanogaster*.

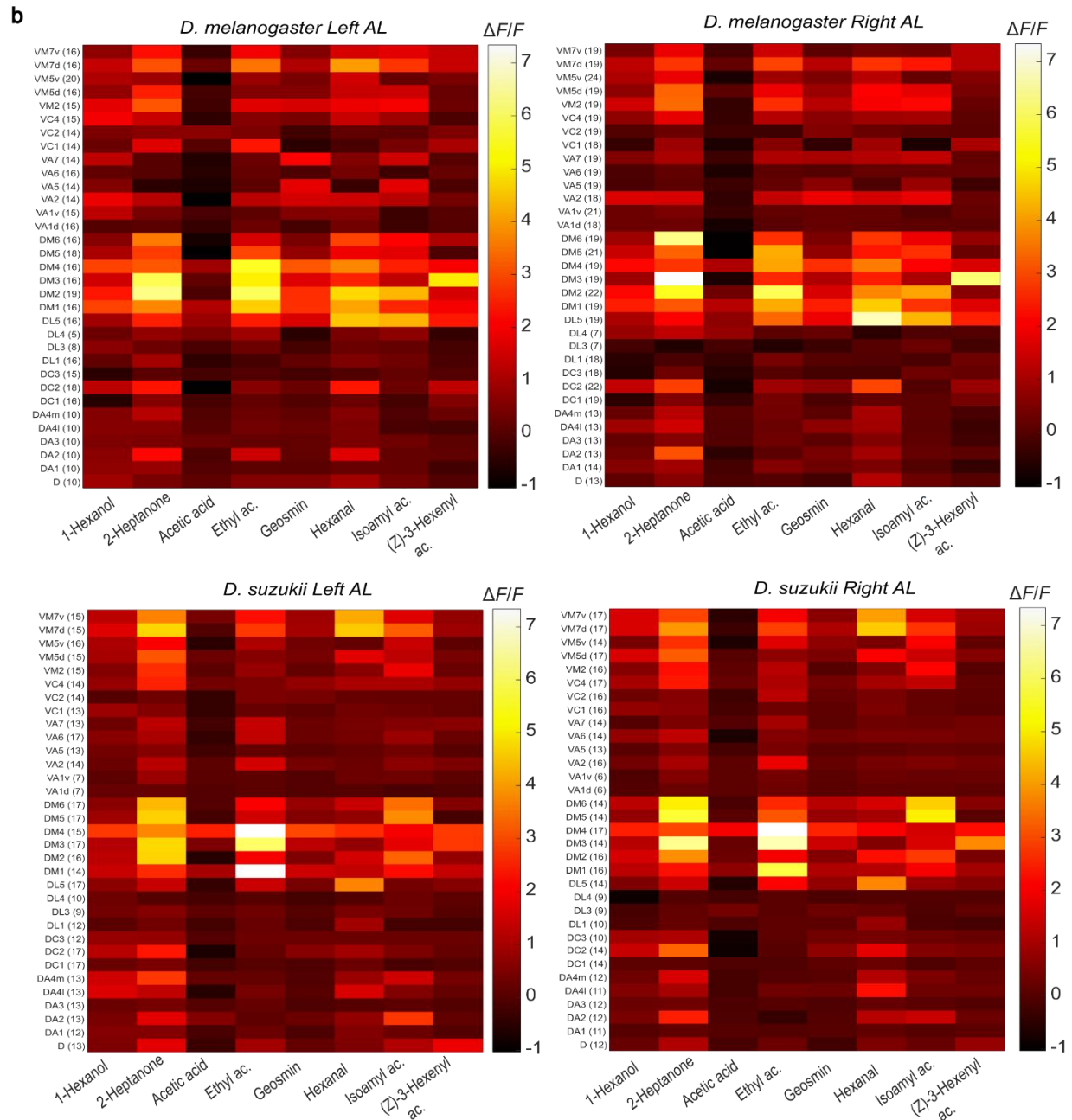
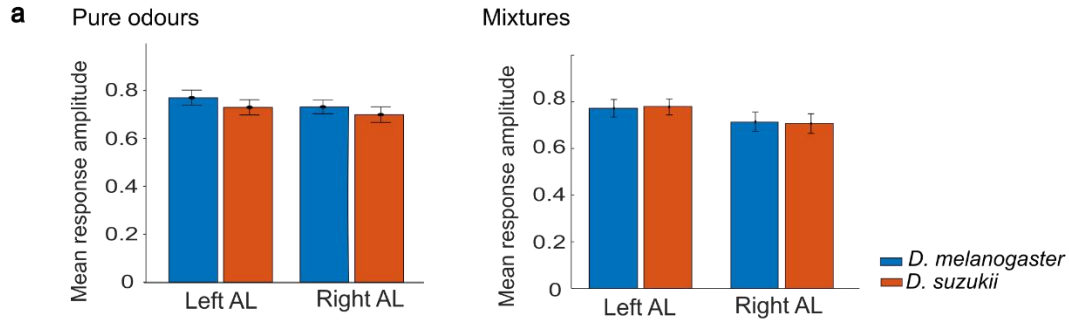
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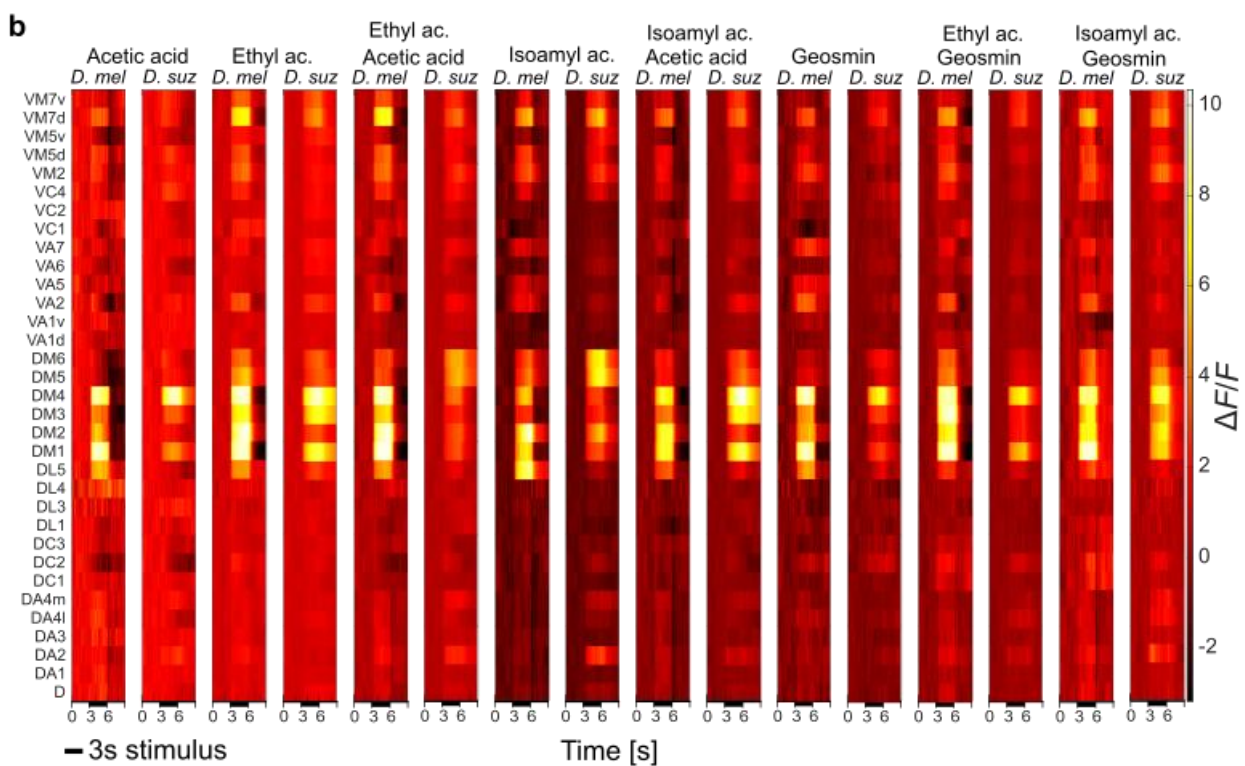
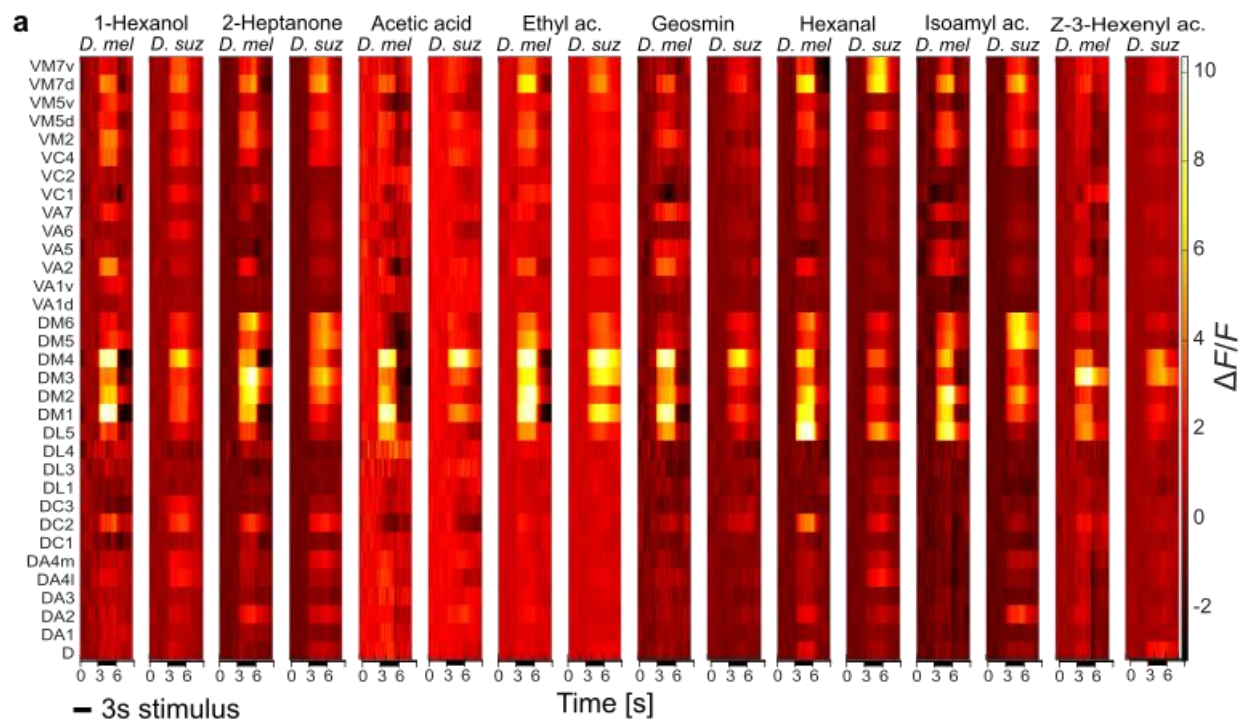
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**Figure S1. Symmetry assessment of odour response patterns in *D. sukuzii* and *D. melanogaster*.**

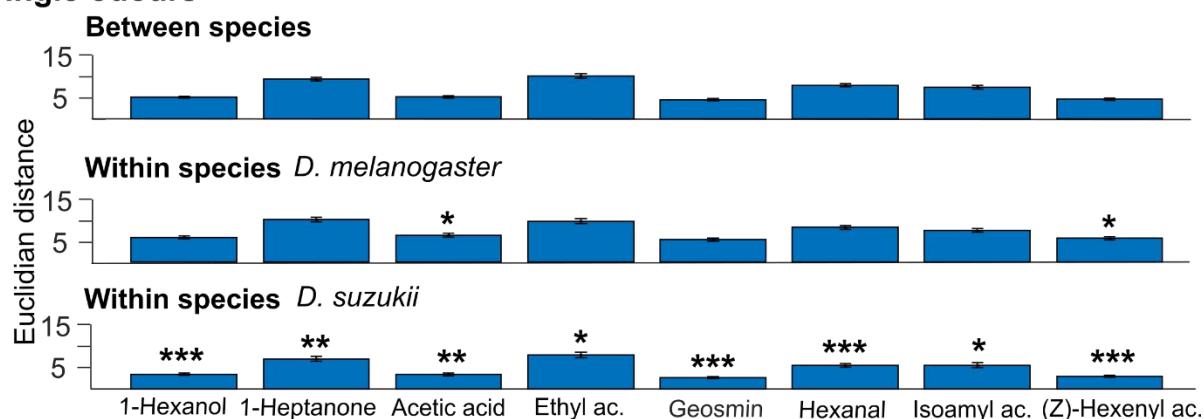
Mean  $\pm$  SEM response amplitude measured in the right and left antennal lobes (AL) averaged over the eight single odours (a) and the four single odours and their mixtures (b) in *D. sukuzii* (orange) and *D. melanogaster* (blue). c) Mean  $\pm$  SEM response amplitude during a 3 s stimulus with eight single odours in each glomerulus on the left and right AL in *D. melanogaster* (upper heatmaps) and *D. sukuzii* (lower heatmaps). The number of replicates is indicated between brackets next to each glomerulus (y-axis). The colour gradient indicates inhibitions (black,  $\Delta F/F < 0$ ) up to the largest activations (white,  $\Delta F/F > 7$ ). ac. = acetate.



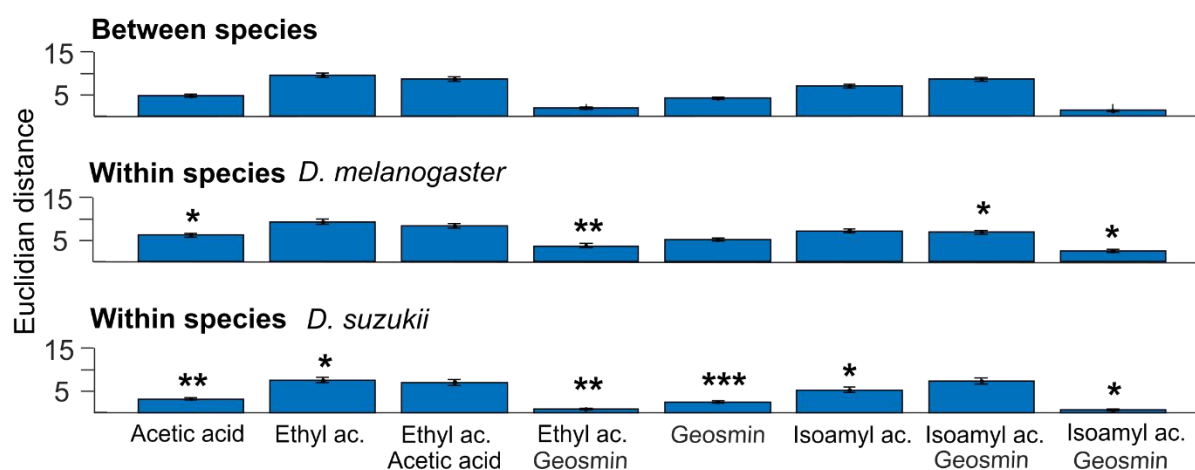
**Figure S2. Odour response maps for eight odorants and their mixtures in *D. suzukii* and *D. melanogaster*.**

Subject-averaged temporal response maps to eight odour compounds before, during, and after a 3 s stimulus plotted for individual glomeruli for eight odours (a) and 4 odours and their mixtures (b).  $\Delta F/F$  was normalised, the same scale was applied to all heatmaps to allow for comparison between species and odours. ‘ac.’ acetate.

## a Single odours



## b Mixtures and components



**Figure S3. Euclidean distance analysis for odour response difference between species vs. within species.**

Mean  $\pm$  SEM of the Euclidean distance (ED) of odour-response amplitudes averaged over all fly pair combinations: between species, within *D. melanogaster*, and within *D. suzukii* in (a) the pure odour experiments and (b) for the mixtures and their components. Significant statistical differences (multiple comparison analysis with FDR correction) in the comparison of between-species ED and within-species ED are labelled according to their significance probabilities as \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ .

**Table S1. 2-way ANOVA of the normalised glomerular volume**

|                    | Sum Sq. | df  | Mean Sq. | F      | p       |
|--------------------|---------|-----|----------|--------|---------|
| <b>Species</b>     | 3.3E-07 | 1   | 3.3E-07  | 0.0017 | 0.97    |
| <b>Glomeruli</b>   | 0.25    | 31  | 0.008    | 42     | 1E-126  |
| <b>Interaction</b> | 0.034   | 31  | 0.0011   | 5.7    | 5.3E-19 |
| <b>Error</b>       | 0.11    | 575 | 1.9E-4   |        |         |
| <b>Total</b>       | 0.39    | 638 |          |        |         |

**Table S2. Multiple comparison analysis of the normalised glomerular volumes**

| Glomerulus  | -95% CI | Mean     | +95% CI | P       | P <sub>FDR</sub> |
|-------------|---------|----------|---------|---------|------------------|
| <b>D</b>    | -0.028  | -0.016   | -0.0038 | 0.01    | 0.046            |
| <b>DA1</b>  | -0.014  | -0.0015  | 0.011   | 0.81    | 0.9              |
| <b>DA2</b>  | -0.009  | 0.0032   | 0.015   | 0.61    | 0.9              |
| <b>DA3</b>  | -0.018  | -0.0058  | 0.0064  | 0.35    | 0.88             |
| <b>DA4L</b> | -0.011  | 0.0015   | 0.014   | 0.81    | 0.9              |
| <b>DA4M</b> | -0.01   | 0.002    | 0.014   | 0.74    | 0.9              |
| <b>DC1</b>  | -0.019  | -0.0065  | 0.0056  | 0.29    | 0.86             |
| <b>DC2</b>  | -0.051  | -0.039   | -0.027  | 5.2E-10 | 8.4E-09          |
| <b>DL1</b>  | -0.014  | -0.0021  | 0.01    | 0.73    | 0.9              |
| <b>DL3</b>  | -0.0078 | 0.0043   | 0.016   | 0.48    | 0.88             |
| <b>DL4</b>  | 0.0093  | 0.021    | 0.034   | 5.5E-4  | 0.0035           |
| <b>DL5</b>  | -0.013  | -0.00083 | 0.011   | 0.89    | 0.91             |
| <b>DM1</b>  | -0.016  | -0.0038  | 0.0083  | 0.53    | 0.88             |
| <b>DM2</b>  | 0.029   | 0.041    | 0.053   | 4.9E-11 | 1.6E-09          |
| <b>DM3</b>  | 0.02    | 0.032    | 0.044   | 3.8E-07 | 4.0E-6           |
| <b>DM4</b>  | -0.016  | -0.0037  | 0.0084  | 0.55    | 0.88             |
| <b>DM5</b>  | -0.027  | -0.015   | -0.0024 | 0.019   | 0.076            |
| <b>DM6</b>  | -0.011  | 0.0012   | 0.013   | 0.84    | 0.9              |

|             |         |         |        |         |        |
|-------------|---------|---------|--------|---------|--------|
| <b>VA1D</b> | -0.0023 | 0.0099  | 0.022  | 0.11    | 0.35   |
| <b>VA1V</b> | -0.014  | -0.0014 | 0.011  | 0.83    | 0.9    |
| <b>VA2</b>  | -0.016  | -0.004  | 0.0082 | 0.52    | 0.88   |
| <b>VA5</b>  | -0.023  | -0.011  | 0.0013 | 0.08    | 0.28   |
| <b>VA6</b>  | -0.0094 | 0.0027  | 0.015  | 0.66    | 0.9    |
| <b>VA7</b>  | -0.0069 | 0.0053  | 0.017  | 0.39    | 0.88   |
| <b>VC1</b>  | -0.017  | -0.0046 | 0.0075 | 0.46    | 0.88   |
| <b>VC2</b>  | -0.015  | -0.0025 | 0.0096 | 0.68    | 0.9    |
| <b>VC3</b>  | -0.011  | 0.0007  | 0.013  | 0.91    | 0.91   |
| <b>VM2</b>  | -0.014  | -0.0015 | 0.011  | 0.81    | 0.9    |
| <b>VM5D</b> | -0.04   | -0.028  | -0.016 | 8.9E-06 | 7.1E-5 |
| <b>VM5V</b> | -0.0067 | 0.0054  | 0.018  | 0.38    | 0.88   |
| <b>VM7D</b> | 0.006   | 0.018   | 0.03   | 0.0035  | 0.019  |
| <b>VM7V</b> | -0.016  | -0.0042 | 0.0079 | 0.5     | 0.88   |

**Table S3. 4-way ANOVA of the mean response amplitudes to pure odours**

| <b>Source</b>             | <b>Sum Sq.</b> | <b>df</b> | <b>Mean Sq.</b> | <b>F</b> | <b>p</b> |
|---------------------------|----------------|-----------|-----------------|----------|----------|
| <b>Species</b>            | 0.0012         | 1         | 0.0012          | 2.8E-4   | 0.99     |
| <b>Side</b>               | 12             | 1         | 12              | 2.8      | 0.094    |
| <b>Odour</b>              | 1800           | 7         | 260             | 60       | 6.5E-86  |
| <b>Glomerulus</b>         | 9400           | 32        | 290             | 67       | 0        |
| <b>Species:side</b>       | 0.13           | 1         | 0.13            | 0.03     | 0.86     |
| <b>Species:odour</b>      | 330            | 7         | 47              | 11       | 1.3E-13  |
| <b>Species:glomerulus</b> | 860            | 32        | 27              | 6.1      | 2.3E-25  |
| <b>Side:odour</b>         | 9.6            | 7         | 1.4             | 0.31     | 0.95     |
| <b>Side:glomerulus</b>    | 230            | 32        | 7.1             | 1.6      | 0.014    |
| <b>Odour:glomerulus</b>   | 3300           | 224       | 15              | 3.3      | 2.4E-55  |
| <b>Error</b>              | 57000          | 13155     | 4.4             |          |          |
| <b>Total</b>              | 75000          | 13499     |                 |          |          |

**Table S4. Multiple comparison analysis of the species dependence for individual glomerular response averaged over the pure odours**

| <b>Glomerulus</b> | <b>-95% CI</b> | <b>Mean</b> | <b>+95% CI</b> | <b><i>P</i></b> | <b><i>P</i><sub>FDR</sub></b> |
|-------------------|----------------|-------------|----------------|-----------------|-------------------------------|
| <b>D</b>          | -0.56          | -0.18       | 0.2            | 0.35            | 0.56                          |
| <b>DA1</b>        | -0.29          | 0.093       | 0.48           | 0.63            | 0.83                          |
| <b>DA2</b>        | -0.46          | -0.086      | 0.29           | 0.66            | 0.83                          |
| <b>DA3</b>        | -0.26          | 0.12        | 0.5            | 0.53            | 0.76                          |
| <b>DA4l</b>       | -0.6           | -0.21       | 0.17           | 0.27            | 0.49                          |
| <b>DA4m</b>       | -0.62          | -0.24       | 0.14           | 0.21            | 0.46                          |
| <b>DC1</b>        | -0.29          | 0.034       | 0.36           | 0.84            | 0.92                          |
| <b>DC2</b>        | -0.15          | 0.17        | 0.48           | 0.3             | 0.49                          |
| <b>DC3</b>        | -0.7           | -0.33       | 0.027          | 0.07            | 0.26                          |
| <b>DL1</b>        | -0.38          | -0.022      | 0.34           | 0.9             | 0.96                          |
| <b>DL3</b>        | -0.65          | -0.19       | 0.26           | 0.41            | 0.61                          |
| <b>DL4</b>        | -0.17          | 0.31        | 0.8            | 0.2             | 0.46                          |
| <b>DL5</b>        | 0.9            | 1.2         | 1.5            | 1.3E-13         | 2.1E-12                       |
| <b>DM1</b>        | 0.24           | 0.57        | 0.89           | 6.6E-4          | 0.0055                        |
| <b>DM2</b>        | 0.88           | 1.2         | 1.5            | 5.2E-14         | 1.7E-12                       |
| <b>DM3</b>        | -0.092         | 0.23        | 0.56           | 0.16            | 0.46                          |
| <b>DM4</b>        | -0.75          | -0.43       | -0.11          | 0.0083          | 0.046                         |
| <b>DM5</b>        | -0.55          | -0.24       | 0.079          | 0.14            | 0.46                          |
| <b>DM6</b>        | -0.53          | -0.21       | 0.11           | 0.21            | 0.46                          |
| <b>VA1d</b>       | -0.48          | -0.05       | 0.38           | 0.82            | 0.92                          |
| <b>VA1v</b>       | -0.42          | 0.005       | 0.43           | 0.98            | 0.98                          |
| <b>VA2</b>        | 0.027          | 0.36        | 0.69           | 0.034           | 0.14                          |
| <b>VA5</b>        | -0.33          | 0.013       | 0.36           | 0.94            | 0.97                          |
| <b>VA6</b>        | -0.69          | -0.37       | -0.044         | 0.026           | 0.12                          |

|             |       |        |       |        |        |
|-------------|-------|--------|-------|--------|--------|
| <b>VA7</b>  | -0.14 | 0.2    | 0.54  | 0.25   | 0.49   |
| <b>VC1</b>  | -0.26 | 0.074  | 0.41  | 0.67   | 0.83   |
| <b>VC2</b>  | -0.4  | -0.07  | 0.26  | 0.68   | 0.83   |
| <b>VC4</b>  | -0.5  | -0.17  | 0.15  | 0.29   | 0.49   |
| <b>VM2</b>  | 0.18  | 0.51   | 0.83  | 0.0023 | 0.015  |
| <b>VM5d</b> | -0.35 | -0.034 | 0.29  | 0.84   | 0.92   |
| <b>VM5v</b> | -0.48 | -0.17  | 0.14  | 0.28   | 0.49   |
| <b>VM7d</b> | -0.54 | -0.22  | 0.096 | 0.17   | 0.46   |
| <b>VM7v</b> | -0.98 | -0.66  | -0.34 | 5.4E-5 | 5.9E-4 |

**Table S5. Multiple comparison analysis of the species dependence for individual pure odour responses averaged over all glomeruli**

| <b>Odour</b>                 | <b>-95% CI</b> | <b>Mean</b> | <b>+95% CI</b> | <b><i>P</i></b> | <b><i>P</i><sub>FDR</sub></b> |
|------------------------------|----------------|-------------|----------------|-----------------|-------------------------------|
| <b>1-Hexanol</b>             | -0.15          | 0.025       | 0.19           | 0.78            | 0.78                          |
| <b>2-Heptanone</b>           | -0.27          | -0.099      | 0.072          | 0.26            | 0.41                          |
| <b>Acetic acid</b>           | -0.33          | -0.16       | 0.015          | 0.074           | 0.2                           |
| <b>Ethyl acetate</b>         | -0.21          | -0.04       | 0.13           | 0.64            | 0.74                          |
| <b>Geosmin</b>               | 0.022          | 0.19        | 0.36           | 0.027           | 0.11                          |
| <b>Hexanal</b>               | 0.16           | 0.33        | 0.5            | 1.4E-4          | 0.0011                        |
| <b>Isoamyl acetate</b>       | -0.25          | -0.084      | 0.087          | 0.34            | 0.45                          |
| <b>(Z)-3-Hexenyl acetate</b> | -0.049         | 0.12        | 0.29           | 0.16            | 0.33                          |

**Table S6. Multiple comparison analysis of Euclidean distances between species vs. within species for pure odours**

|                              | Between species | <i>D. melanogaster</i> |          |                         | <i>D. suzukii</i> |          |                         |
|------------------------------|-----------------|------------------------|----------|-------------------------|-------------------|----------|-------------------------|
|                              | Mean ± SEM      | Mean ± SEM             | <i>P</i> | <i>P</i> <sub>FDR</sub> | Mean ± SEM        | <i>P</i> | <i>P</i> <sub>FDR</sub> |
| <b>1Hexanol</b>              | 4.8 ± 0.22      | 5.5 ± 0.31             | 0.054    | 0.072                   | 3.2 ± 0.25        | 6.10E-06 | <b>3.20E-05</b>         |
| <b>2-Heptanone</b>           | 8.9 ± 0.42      | 9.5 ± 0.54             | 0.34     | 0.42                    | 6.7 ± 0.57        | 0.0024   | <b>0.0064</b>           |
| <b>Acetic acid</b>           | 4.9 ± 0.29      | 6 ± 0.46               | 0.025    | 0.04                    | 3.2 ± 0.31        | 4.10E-04 | <b>0.0013</b>           |
| <b>Ethyl acetate</b>         | 9.6 ± 0.47      | 9.2 ± 0.57             | 0.6      | 0.65                    | 7.6 ± 0.66        | 0.013    | <b>0.026</b>            |
| <b>Geosmin</b>               | 4.3 ± 0.23      | 5 ± 0.36               | 0.053    | 0.072                   | 2.5 ± 0.21        | 7.80E-07 | <b>1.20E-05</b>         |
| <b>Hexanal</b>               | 7.5 ± 0.35      | 7.8 ± 0.47             | 0.6      | 0.65                    | 5.2 ± 0.44        | 1.00E-04 | <b>4.10E-04</b>         |
| <b>Isoamyl acetate</b>       | 7.1 ± 0.41      | 7.1 ± 0.42             | 0.92     | 0.92                    | 5.3 ± 0.59        | 0.013    | <b>0.026</b>            |
| <b>(Z)-3-Hexenyl acetate</b> | 4.4 ± 0.23      | 5.4 ± 0.36             | 0.015    | 0.027                   | 2.8 ± 0.22        | 5.90E-06 | <b>3.20E-05</b>         |

**Table S7. 4-way ANOVA of mean response amplitudes for mixed odours and components**

|                           | Sum Sq. | <i>df</i> | Mean Sq. | <i>F</i> | <i>p</i>       |
|---------------------------|---------|-----------|----------|----------|----------------|
| <b>Species</b>            | 0.0012  | 1         | 0.0012   | 2.8E-4   | 0.99           |
| <b>Side</b>               | 12      | 1         | 12       | 2.8      | 0.094          |
| <b>Odour</b>              | 1800    | 7         | 260      | 60       | <b>6.5E-86</b> |
| <b>Glomerulus</b>         | 9400    | 32        | 290      | 0        | 1              |
| <b>Species:side</b>       | 0.13    | 1         | 0.13     | 0.03     | 0.86           |
| <b>Species:odour</b>      | 330     | 7         | 47       | 11       | <b>1.3E-13</b> |
| <b>Species:glomerulus</b> | 860     | 32        | 27       | 6.1      | <b>2.3E-25</b> |
| <b>Side:odour</b>         | 9.6     | 7         | 1.4      | 0.31     | 0.95           |
| <b>Side:glomerulus</b>    | 230     | 32        | 7.1      | 1.6      | <b>0.014</b>   |
| <b>Odour:glomerulus</b>   | 3300    | 224       | 15       | 3.3      | <b>2.4E-55</b> |
| <b>Error</b>              | 57000   | 13155     |          |          |                |
| <b>Total</b>              | 75000   | 13499     |          |          |                |

**Table S8. Multiple comparison analysis of the species dependence for individual glomerular responses averaged over the mixed odours and their components**

| <b>Glomerulus</b> | <b>-95% CI</b> | <b>Mean</b> | <b>+95% CI</b> | <b><i>P</i></b> | <b><i>P</i><sub>FDR</sub></b> |
|-------------------|----------------|-------------|----------------|-----------------|-------------------------------|
| <b>D</b>          | -0.47          | -0.012      | 0.44           | 0.96            | 0.96                          |
| <b>DA1</b>        | -0.33          | 0.13        | 0.59           | 0.58            | 0.77                          |
| <b>DA2</b>        | -0.77          | -0.32       | 0.14           | 0.17            | 0.48                          |
| <b>DA3</b>        | -0.33          | 0.12        | 0.57           | 0.61            | 0.77                          |
| <b>DA4l</b>       | -0.47          | -0.013      | 0.44           | 0.95            | 0.96                          |
| <b>DA4m</b>       | -0.65          | -0.2        | 0.25           | 0.38            | 0.74                          |
| <b>DC1</b>        | -0.26          | 0.14        | 0.53           | 0.5             | 0.74                          |
| <b>DC2</b>        | -0.36          | 0.023       | 0.4            | 0.91            | 0.96                          |
| <b>DC3</b>        | -0.3           | 0.14        | 0.57           | 0.54            | 0.74                          |
| <b>DL1</b>        | -0.53          | -0.092      | 0.34           | 0.68            | 0.83                          |
| <b>DL3</b>        | -0.75          | -0.21       | 0.33           | 0.45            | 0.74                          |
| <b>DL4</b>        | -0.35          | 0.23        | 0.8            | 0.44            | 0.74                          |
| <b>DL5</b>        | 0.97           | 1.4         | 1.8            | 1.2E-11         | 2E-10                         |
| <b>DM1</b>        | -0.24          | 0.14        | 0.53           | 0.46            | 0.74                          |
| <b>DM2</b>        | 1.2            | 1.6         | 2              | 2.9E-17         | 9.5E-16                       |
| <b>DM3</b>        | -0.85          | -0.46       | -0.066         | 0.022           | 0.1                           |
| <b>DM4</b>        | -1.1           | -0.71       | -0.33          | 2.5E-4          | 0.0019                        |
| <b>DM5</b>        | -1.1           | -0.74       | -0.35          | 1.6E-4          | 0.0018                        |
| <b>DM6</b>        | -1.1           | -0.73       | -0.33          | 2.9E-4          | 0.0019                        |
| <b>VA1d</b>       | -0.59          | -0.086      | 0.41           | 0.74            | 0.87                          |
| <b>VA1v</b>       | -0.83          | -0.33       | 0.17           | 0.19            | 0.49                          |
| <b>VA2</b>        | -0.37          | 0.023       | 0.42           | 0.91            | 0.96                          |
| <b>VA5</b>        | -0.16          | 0.25        | 0.66           | 0.24            | 0.52                          |
| <b>VA6</b>        | -0.76          | -0.36       | 0.032          | 0.072           | 0.26                          |
| <b>VA7</b>        | -0.28          | 0.13        | 0.53           | 0.54            | 0.74                          |

|             |        |       |         |        |       |
|-------------|--------|-------|---------|--------|-------|
| <b>VC1</b>  | -0.16  | 0.24  | 0.64    | 0.23   | 0.52  |
| <b>VC2</b>  | -0.52  | -0.13 | 0.26    | 0.51   | 0.74  |
| <b>VC4</b>  | -0.39  | -0.01 | 0.37    | 0.96   | 0.96  |
| <b>VM2</b>  | -0.074 | 0.31  | 0.7     | 0.11   | 0.37  |
| <b>VM5d</b> | -0.12  | 0.26  | 0.64    | 0.18   | 0.48  |
| <b>VM5v</b> | -0.92  | -0.54 | -0.16   | 0.0051 | 0.028 |
| <b>VM7d</b> | -0.17  | 0.2   | 0.58    | 0.29   | 0.6   |
| <b>VM7v</b> | -0.76  | -0.38 | -0.0042 | 0.047  | 0.2   |

**Table S9. Multiple comparison analysis of the species dependence for individual mixed odours and components averaged over all glomeruli**

| <b>Odour</b>                         | <b>-95% CI</b> | <b>Mean</b> | <b>+95% CI</b> | <b><i>P</i></b> | <b><i>P</i><sub>FDR</sub></b> |
|--------------------------------------|----------------|-------------|----------------|-----------------|-------------------------------|
| <b>Acetic acid</b>                   | -0.34          | -0.15       | 0.037          | 0.12            | 0.23                          |
| <b>Ethyl acetate</b>                 | -0.22          | -0.036      | 0.15           | 0.71            | 0.81                          |
| <b>Ethyl acetate - Acetic acid</b>   | -0.18          | 0.0096      | 0.2            | 0.92            | 0.92                          |
| <b>Ethyl acetate - Geosmin</b>       | 0.36           | 0.64        | 0.93           | 9.7E-06         | 3.9E-05                       |
| <b>Geosmin</b>                       | 0.0072         | 0.2         | 0.38           | 0.042           | 0.11                          |
| <b>Isoamyl acetate</b>               | -0.27          | -0.08       | 0.11           | 0.41            | 0.65                          |
| <b>Isoamyl acetate - Acetic acid</b> | -0.87          | -0.68       | -0.49          | 1.4E-12         | 1.1E-11                       |
| <b>Isoamyl acetate - Geosmin</b>     | -0.19          | 0.098       | 0.38           | 0.5             | 0.67                          |

**Table S10. Multiple comparison analysis of Euclidean distances between species vs. within species for odour mixtures and their components**

|                                      | Between species<br>Mean $\pm$ SEM | <i>D. melanogaster</i> |          |                         | <i>D. suzukii</i> |          |                         |
|--------------------------------------|-----------------------------------|------------------------|----------|-------------------------|-------------------|----------|-------------------------|
|                                      |                                   | Mean $\pm$ SEM         | <i>P</i> | <i>P</i> <sub>FDR</sub> | Mean $\pm$ SEM    | <i>P</i> | <i>P</i> <sub>FDR</sub> |
| <b>Acetic acid</b>                   | 4.9 $\pm$ 0.29                    | 6 $\pm$ 0.46           | 0.025    | <b>0.04</b>             | 3.2 $\pm$ 0.31    | 4.10E-04 | <b>3.30E-03</b>         |
| <b>Ethyl acetate</b>                 | 9.6 $\pm$ 0.47                    | 9.2 $\pm$ 0.57         | 0.6      | 0.64                    | 7.6 $\pm$ 0.66    | 0.013    | <b>0.023</b>            |
| <b>Ethyl acetate - Acetic acid</b>   | 8.7 $\pm$ 0.48                    | 8.2 $\pm$ 0.51         | 0.52     | 0.6                     | 7.1 $\pm$ 0.71    | 5.70E-02 | 0.076                   |
| <b>Ethyl acetate - Geosmin</b>       | 2 $\pm$ 0.26                      | 3.6 $\pm$ 0.52         | 0.0025   | <b>0.0099</b>           | 0.82 $\pm$ 0.19   | 0.0016   | <b>0.0084</b>           |
| <b>Geosmin</b>                       | 4.3 $\pm$ 0.23                    | 5 $\pm$ 0.36           | 0.053    | 0.076                   | 2.5 $\pm$ 0.21    | 7.80E-07 | <b>1.20E-05</b>         |
| <b>Isoamyl acetate</b>               | 7.1 $\pm$ 0.41                    | 7.1 $\pm$ 0.42         | 0.92     | 0.92                    | 5.3 $\pm$ 0.59    | 1.30E-02 | <b>2.30E-02</b>         |
| <b>Isoamyl acetate - Acetic acid</b> | 8.7 $\pm$ 0.46                    | 6.8 $\pm$ 0.39         | 0.0065   | <b>0.015</b>            | 7.4 $\pm$ 0.69    | 0.1      | 0.13                    |
| <b>Isoamyl acetate - Geosmin</b>     | 1.4 $\pm$ 0.18                    | 2.5 $\pm$ 0.38         | 0.0033   | <b>0.01</b>             | 0.62 $\pm$ 0.15   | 5.60E-03 | <b>1.50E-02</b>         |

**Table S11. 2-way ANOVA of the odour- and species-dependence for 3 behavioural experiments each with one different reference odour (bait1)**

**Bait1 = Acetic acid**

|                      | <b>Sum Sq.</b> | <b>df</b> | <b>Mean Sq.</b> | <b>F</b> | <b>p</b>       |
|----------------------|----------------|-----------|-----------------|----------|----------------|
| <b>bait2</b>         | 0.83           | 2         | 0.42            | 36       | <b>6.3E-11</b> |
| <b>species</b>       | 0.000074       | 1         | 0.000074        | 0.0065   | 0.94           |
| <b>bait2:species</b> | 0.0064         | 2         | 0.0032          | 0.28     | 0.76           |
| <b>Error</b>         | 0.67           | 58        | 0.012           |          |                |
| <b>Total</b>         | 1.5            | 63        |                 |          |                |

**Bait1 = Ethyl acetate**

|                      | <b>Sum Sq.</b> | <b>df</b> | <b>Mean Sq.</b> | <b>F</b> | <b>p</b>       |
|----------------------|----------------|-----------|-----------------|----------|----------------|
| <b>bait2</b>         | 3.7            | 4         | 0.91            | 77       | <b>4.1E-38</b> |
| <b>species</b>       | 0.01           | 1         | 0.01            | 0.86     | 0.35           |
| <b>bait2:species</b> | 0.2            | 4         | 0.051           | 4.3      | <b>0.0025</b>  |
| <b>Error</b>         | 2.2            | 182       | 0.012           |          |                |
| <b>Total</b>         | 6              | 191       |                 |          |                |

**Bait1 = Isoamyl acetate**

|                      | <b>Sum Sq.</b> | <b>df</b> | <b>Mean Sq.</b> | <b>F</b> | <b>p</b>       |
|----------------------|----------------|-----------|-----------------|----------|----------------|
| <b>bait2</b>         | 2.3            | 4         | 0.58            | 30       | <b>2.8E-19</b> |
| <b>species</b>       | 0.0029         | 1         | 0.0029          | 0.15     | 0.7            |
| <b>bait2:species</b> | 0.073          | 4         | 0.018           | 0.94     | 0.44           |
| <b>Error</b>         | 3.5            | 182       | 0.019           |          |                |
| <b>Total</b>         | 5.9            | 191       |                 |          |                |

**Table S12. Multiple comparisons of odour- and species-dependence for 3 behavioural experiments each with a different reference odour (Bait1)**

| <i>D. suzukii</i> vs <i>D. melanogaster</i> | -95% CI  | mean    | +95% CI | <i>P</i> | <i>P<sub>FDR</sub></i> |
|---------------------------------------------|----------|---------|---------|----------|------------------------|
| <b>Bait1 = Acetic acid</b>                  |          |         |         |          |                        |
| Acetic acid                                 | -0.083   | -0.0068 | 0.069   | 0.86     | 0.91                   |
| Empty                                       | -0.074   | 0.033   | 0.14    | 0.54     | 0.78                   |
| Paraffin oil                                | -0.13    | -0.02   | 0.088   | 0.72     | 0.85                   |
| <b>Bait1 = Ethyl acetate</b>                |          |         |         |          |                        |
| Empty                                       | -0.098   | -0.036  | 0.026   | 0.26     | 0.48                   |
| Ethyl acetate                               | -0.00032 | 0.053   | 0.11    | 0.051    | 0.33                   |
| Paraffin oil                                | -0.15    | -0.088  | -0.026  | 0.0059   | 0.076                  |
| Acetic acid                                 | -0.021   | 0.087   | 0.19    | 0.11     | 0.48                   |
| Ethyl acetate + Acetic acid                 | -0.038   | 0.07    | 0.18    | 0.2      | 0.48                   |
| <b>Bait1 = Isoamyl acetate</b>              |          |         |         |          |                        |
| Acetic acid                                 | -0.054   | 0.083   | 0.22    | 0.24     | 0.48                   |
| Empty                                       | -0.075   | 0.0044  | 0.083   | 0.91     | 0.91                   |
| Isoamyl acetate                             | -0.11    | -0.041  | 0.028   | 0.24     | 0.48                   |
| Paraffin oil                                | -0.045   | 0.034   | 0.11    | 0.4      | 0.65                   |
| Isoamyl acetate + Acetic acid               | -0.17    | -0.034  | 0.1     | 0.62     | 0.81                   |

**Table S13 Within-species multiple comparisons of behavioural odour preferences.**  
**Comparisons with controls are not shown as odours were always significantly preferred over paraffin oil, which in turn was significantly preferred over the empty vial.**

**Bait1 = Ethyl acetate**

| <b>Bait2</b>                   | <b>Bait2</b>                  | <b>-95% CI</b> | <b>mean</b> | <b>+95% CI</b> | <b>p</b> | <b>p<sub>FDR</sub></b> |
|--------------------------------|-------------------------------|----------------|-------------|----------------|----------|------------------------|
| <b><i>D. melanogaster</i></b>  |                               |                |             |                |          |                        |
| Ethyl acetate                  | Acetic acid                   | -0.076         | 0.0092      | 0.094          | 0.83     | 0.83                   |
| Ethyl acetate                  | Ethyl acetate + Acetic acid   | -0.21          | -0.12       | -0.036         | 0.0056   | 0.0081                 |
| Acetic acid                    | Ethyl acetate + Acetic acid   | -0.24          | -0.13       | -0.022         | 0.018    | 0.02                   |
| <b><i>D. suzukii</i></b>       |                               |                |             |                |          |                        |
| Ethyl acetate                  | Acetic acid                   | -0.043         | 0.042       | 0.13           | 0.33     | 0.33                   |
| Ethyl acetate                  | Ethyl acetate + Acetic acid   | -0.19          | -0.1        | -0.019         | 0.016    | 0.02                   |
| Acetic acid                    | Ethyl acetate + Acetic acid   | -0.25          | -0.15       | -0.039         | 0.0077   | 0.011                  |
| <b>Bait1 = Isoamyl acetate</b> |                               |                |             |                |          |                        |
| <b>Bait2</b>                   | <b>Bait2</b>                  | <b>-95% CI</b> | <b>mean</b> | <b>+95% CI</b> | <b>p</b> | <b>p<sub>FDR</sub></b> |
| <b><i>D. melanogaster</i></b>  |                               |                |             |                |          |                        |
| Acetic acid                    | Isoamyl acetate               | 0.014          | 0.12        | 0.23           | 0.027    | 0.034                  |
| Acetic acid                    | Isoamyl acetate + Acetic acid | -0.17          | -0.038      | 0.099          | 0.59     | 0.59                   |
| Isoamyl acetate                | Isoamyl acetate + Acetic acid | -0.27          | -0.16       | -0.052         | 0.004    | 0.0067                 |
| <b><i>D. suzukii</i></b>       |                               |                |             |                |          |                        |
| Acetic acid                    | Isoamyl acetate               | 0.014          | 0.25        | 0.36           | 1.4E-5   | 3.4E-5                 |
| Acetic acid                    | Isoamyl acetate + Acetic acid | -0.058         | 0.079       | 0.22           | 0.26     | 0.27                   |
| Isoamyl acetate                | Isoamyl acetate + Acetic acid | -0.28          | -0.17       | -0.059         | 0.0028   | 0.004                  |